

Abhishek Singh

B.Tech Electronics and Communication Engineering, IIT Patna (Batch of 2027)

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Education

Indian Institute of Technology Patna

B.Tech in Electronics and Communication Engineering (Electrical Department)

2023–2027

CPI: 7.0/10.0

Scholastic Achievements

- **AIR 1046 in GATE (ECE)**, Score 662.
- AIR 5701 in JEE Advanced and AIR 4311 in JEE Main.
- Solved 1000+ DSA/algorithm problems across competitive programming platforms.

Hardware & Chip-Relevant Research Experience

Research under Prof. Sumanta Gupta, IIT Patna

2026–Present

Photonics / Optical Communication Group

- Exploring photonic-electronic implementations of neural-network primitives, focusing on optical-domain multiply-accumulate (MAC) operations for **silicon photonic hardware**.
- Studying microring-resonator weighting and balanced-detection schemes for signed-weight optical computation — device-level analog signal design.
- Working toward a photoelectronic multiply-accumulate neuron (PEMAN) architecture combining optical-domain multiplication with electronic accumulation and activation.

Independent Research — Silicon Photonic Microring Resonators

2026

- Simulated lateral, vertical/3D, and parallel-cascaded coupling configurations of on-chip resonator structures using the open-source MEEP FDTD electromagnetic simulation framework.
- Performed coupling-gap sweeps, critical-coupling analysis, and quality-factor (Q) extraction via coupled-mode theory — device-characterization methodology transferable to on-chip component design.
- Authored a citable technical report (Zenodo).

Software Head, Inditronix AI Labs

2024–Present

- Built ESP32-based IoT hardware applications, including sensor interfacing, embedded firmware, and low-power embedded system design; developed supporting infrastructure on AWS EC2.

Projects

Photonic-Electronic MAC Neuron (PEMAN) — Ongoing

- Designing a hybrid architecture combining optical-domain multiplication (microring weighting) with electronic accumulation/activation for neural-network hardware acceleration.

ESP32-based IoT Applications

- Designed embedded firmware and sensor-interfacing circuits for healthcare IoT devices; coordinated hardware bring-up and validation.

Technical Skills

EDA / Simulation: MEEP (FDTD), coupled-mode theory analysis **Hardware:** ESP32 embedded systems, sensor interfacing

Languages: Python, C++, C, SQL **Domains:** VLSI/digital design, semiconductor devices, RF/analog systems, DSP, embedded systems

Tools: Linux, Git, NumPy, LaTeX

Affiliations

IEEE Student Member — IEEE Photonics Society (since June 2026).